

Project 6: Application Load Balancer + Auto Scaling Group (High Availability)

Objective

To balance traffic across multiple EC2 instances and automatically scale them based on traffic demand to ensure the website remains highly available and always online.

Step-by-Step Implementation Guide

Step 1: Create a Launch Template

1. Open the **EC2 Console** → Click on **Launch Templates** in the left menu → Click **Create template**.
2. **Template Name:** Enter my-web-server-template.
3. **AMI:** Select *Amazon Linux 2023*.
4. **Instance Type:** Choose *t2.micro*.
5. **Key Pair:** Select your existing key pair or create a new one.
6. Expand the **Advanced Details** section, scroll down to the **User Data** field, and paste the following script (this script automatically installs and configures the Apache web server):

Bash

```
#!/bin/bash
```

```
sudo yum update -y
```

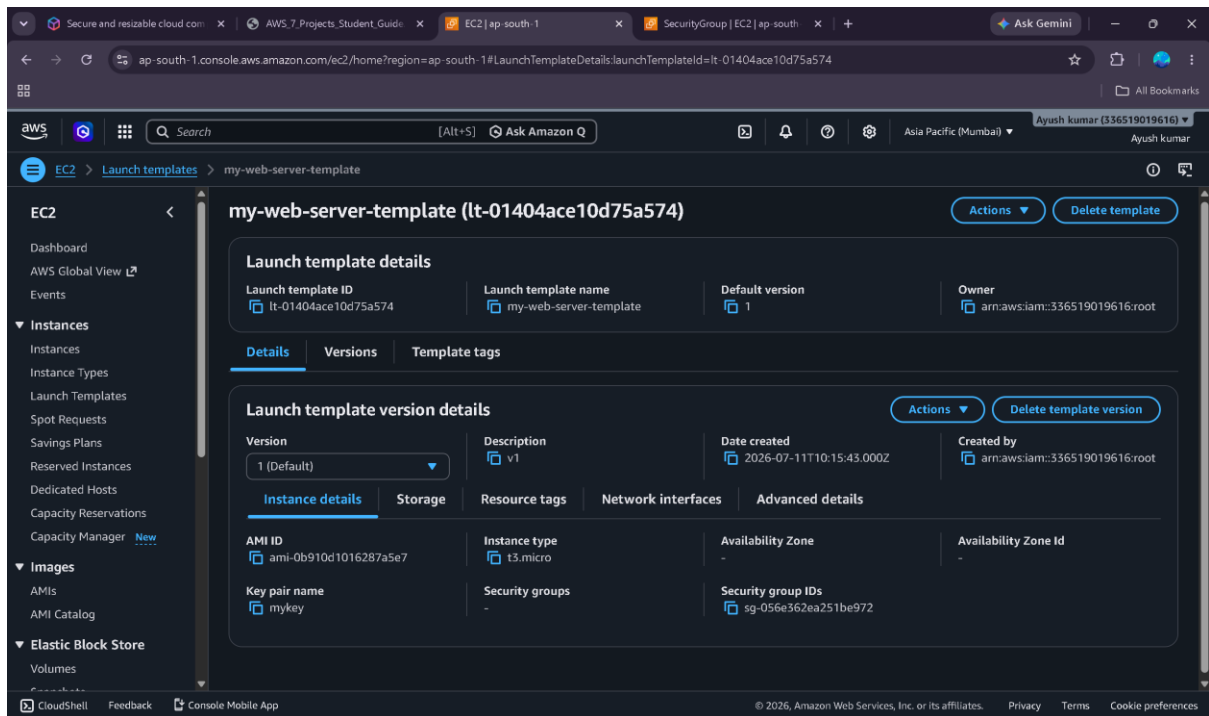
```
sudo yum install httpd -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

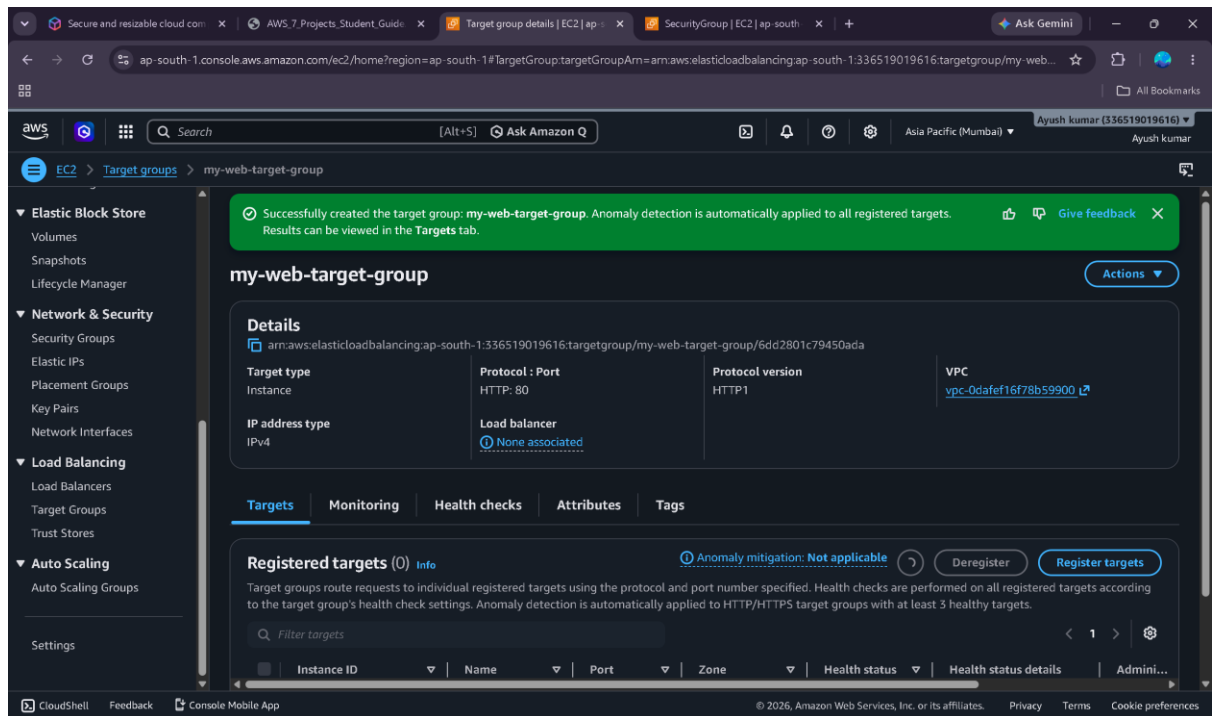
```
echo "<h1>Server: $(hostname -f)</h1>" > /var/www/html/index.html
```

7. Click **Create launch template**.



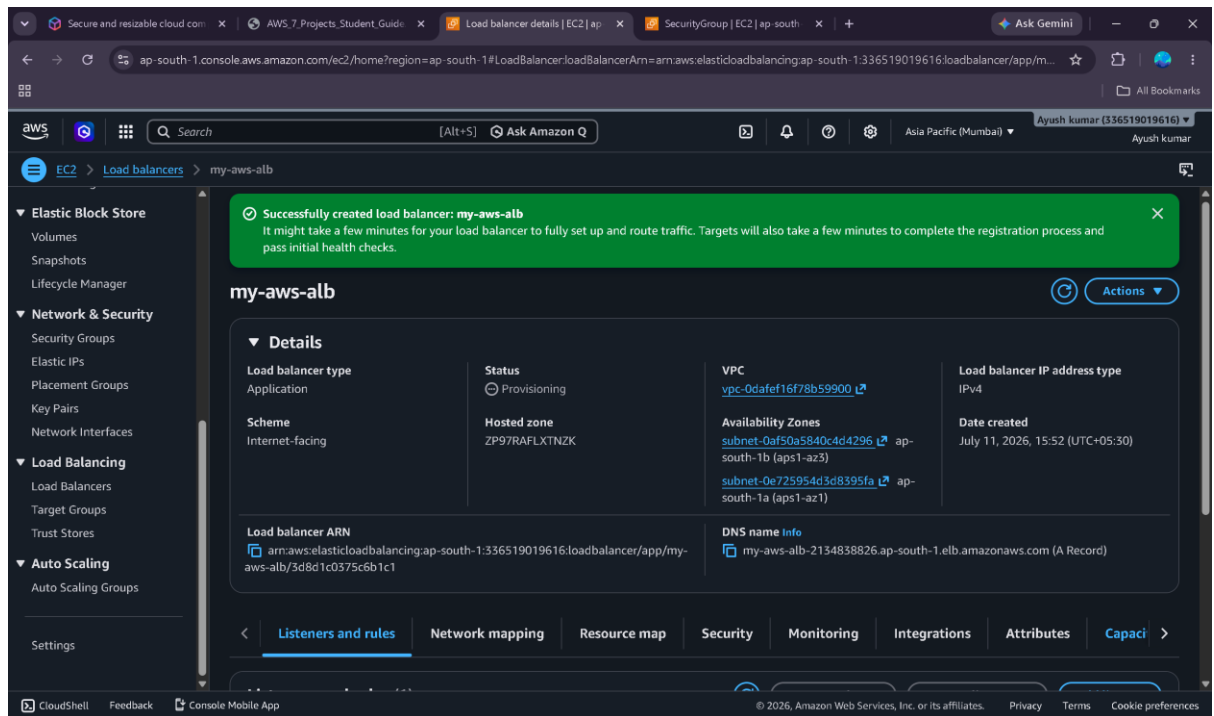
Step 2: Create a Target Group

1. In the EC2 console left menu, click on **Target Groups** → **Create target group**.
2. **Choose a target type:** Select *Instances*.
3. **Target Group Name:** Enter my-web-target-group.
4. Keep **Protocol** as *HTTP* and **Port** as *80*.
5. **VPC:** Select your default VPC or the specific VPC you are using.
6. **Health check path:** Leave it as */* (default).
7. Click **Next**, and without registering any instances manually at this stage, click **Create target group** (the Auto Scaling Group will automatically manage and add instances here).



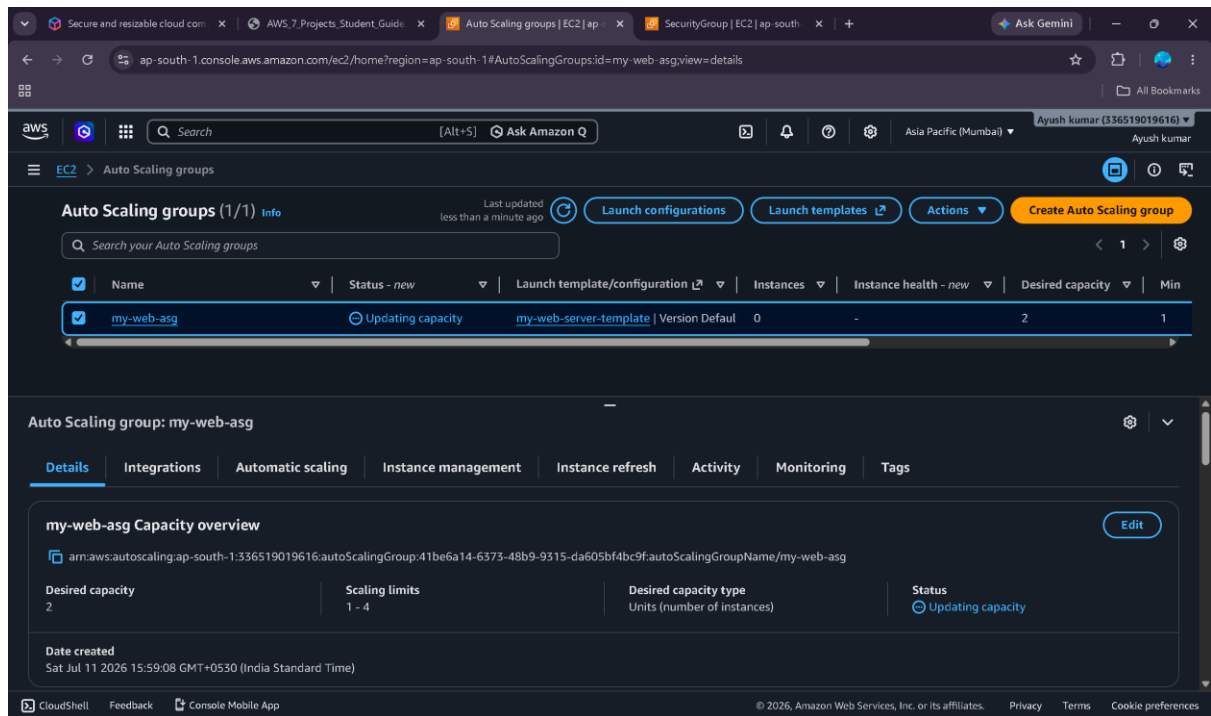
Step 3: Create an Application Load Balancer (ALB)

1. In the EC2 console, click on **Load Balancers** → **Create load balancer**.
2. Under **Load Balancer Type**, select **Application Load Balancer** and click **Create**.
3. **Load Balancer Name**: Enter my-aws-alb.
4. **Scheme**: Select *Internet-facing*.
5. **Network Mapping**: Select your VPC and check/select public subnets from at least **2 Availability Zones (AZs)** to ensure High Availability.
6. **Security Groups**: Choose a Security Group that allows public inbound traffic on port 80 (HTTP).
7. **Listeners and Routing**: * **Protocol**: *HTTP*, **Port**: *80*
 - **Default Action**: Select the Target Group you created (my-web-target-group).
8. Click **Create load balancer**.



Step 4: Create an Auto Scaling Group (ASG)

1. In the EC2 console, click on **Auto Scaling Groups** → **Create Auto Scaling group**.
2. **Name:** Enter my-web-asg.
3. **Launch Template:** Select the template created in Step 1 → Click **Next**.
4. **Network:** Select your VPC and choose multiple subnets (across different AZs) where you want the instances to be launched → Click **Next**.
5. **Configure advanced options:** * **Load Balancing:** Choose *Attach to an existing load balancer*.
 - **Target Group:** Select your target group (my-web-target-group).
 - **Health checks:** You can optionally check *ELB* → Click **Next**.
6. **Configure group size and scaling policies:**
 - **Desired capacity:** 2
 - **Minimum capacity:** 1
 - **Maximum capacity:** 4
 - **Scaling Policies:** Select *Target tracking scaling policy*.
 - **Metric type:** Select *Average CPU utilization*, and set the **Target value** to 50 → Click **Next**.
7. Keep the remaining settings as default and click **Create Auto Scaling group**.



Live Verification & Testing

1. Verify Load Balancing

- Go to the **Load Balancers** tab, select your ALB, and copy its **DNS name**.
- Paste the DNS name into a web browser tab and hit enter. You will see a web page displaying the hostname of an instance (e.g., Server: ip-10-0-x-x.ap-south-1.compute.internal).
- Keep **Refreshing** the browser tab. You will notice the hostname changes between your running instances, proving that the ALB is successfully distributing traffic across them.

Server: ip-10-0-2-4.ap-south-1.compute.internal

2. Verify Auto Scaling & Self-Healing

- Navigate to the **EC2 Instances** list. You will see the 2 instances launched automatically by the ASG.
- Select **one of the instances** and manually **Terminate it**.
- Now, go to your **Auto Scaling Group** and click on the **Activity** tab. You will observe that the ASG automatically detects the unhealthy/missing instance and immediately **launches a new instance** to maintain the desired capacity. This demonstrates the **Self-Healing** capabilities of AWS.

